

Calculus A

Chapter 2: Limits and Continuity

Section 2.4: One-Sided Limits

Main Lecture Notes

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Reference: Thomas' Calculus, 12th Edition

One-Sided Limits

Fact

Sometimes saying that the limit of a function fails to exist at a point does not provide us with enough information about the behavior of the function at that particular point. To see this, let us revisit the function

$$g(x) = \frac{|x-2|}{(x-2)}$$

introduced in the previous section (Sec2.2). As we choose values of x close to 2, $g(x)$ does not approach a single value, so the limit as x approaches 2 does not exist-that is, $\lim_{x\to 2} g(x)$ DNE. However, this statement alone does not give us a complete picture of the behavior of the function around the $x-$value 2. To provide a more accurate description, we introduce the idea of a **one-sided limit**.
For all values to the left of 2 (or the negative side of 2), $g(x) = -1$. Thus, as x approaches 2 from the left, $g(x)$ approaches -1.

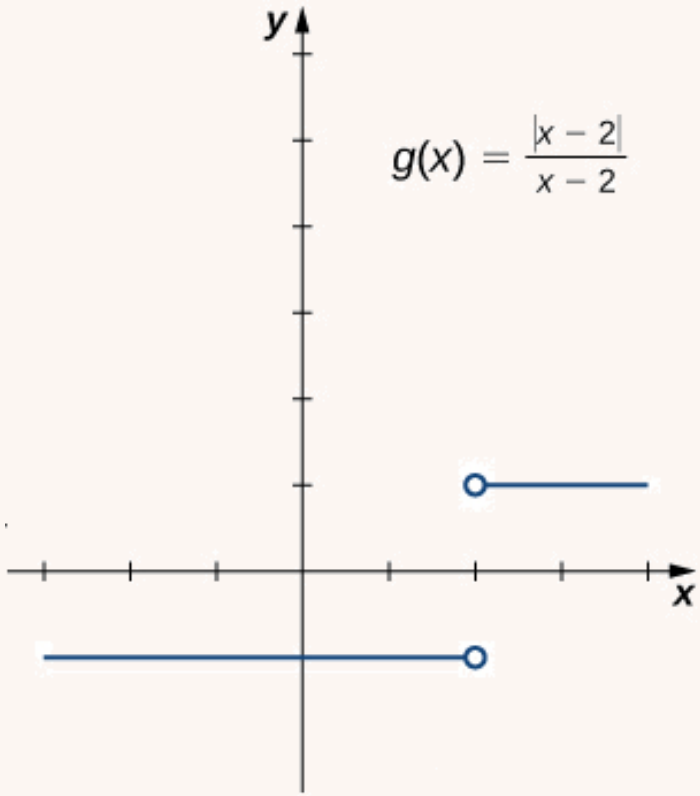
Fact Cont..

Mathematically, we say that the limit as x approaches 2 from the left is -1. Symbolically, we express this idea as

$$\lim_{x \to 2^-} g(x) = -1.$$

Similarly, as x approaches 2 from the right (or from the positive side), $g(x)$ approaches 1. Symbolically, we express this idea as

$$\lim_{x \to 2^+} g(x) = 1.$$



We can now present an informal definition of one-sided limits.

One-Sided Limits

Informal Definition of One-Sided Limit

We define two types of one-sided limits.

Limit from the left:

Let $f(x)$ be a function defined at all values in an open interval (a, c), where $a < c$, and let L be a real number.

If the values of the function $f(x)$ approach the real number L as the values of x (where $x < c$) approach the number c, then we say that L is the limit of $f(x)$ as x approaches c from the left.

Symbolically, we express this idea as

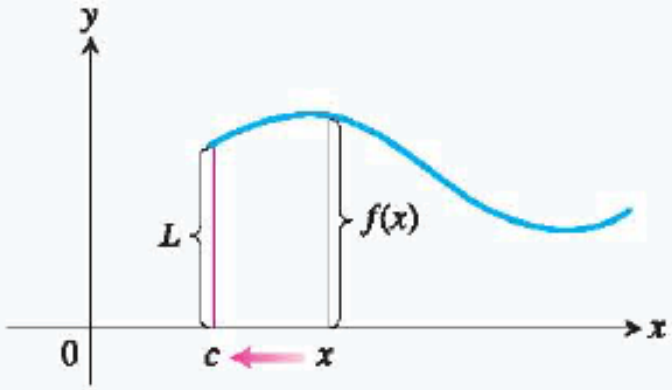
$$\lim_{x \to c^-} f(x) = L.$$

Informal Definition of One-Sided Limit Cont..

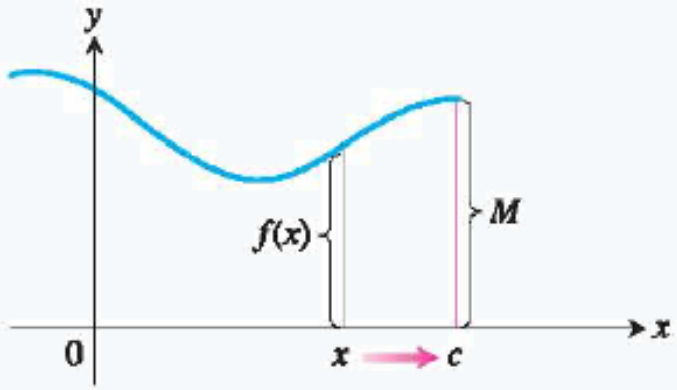
Limit from the right:

Let $f(x)$ be a function defined at all values in an open interval (c, b), where $c < b$, and let M be a real number. If the values of the function $f(x)$ approach the real number M as the values of x (where $x > c$) approach the number c, then we say that M is the limit of $f(x)$ as x approaches c from the right. Symbolically, we express this idea as

$$\lim_{x \to c^+} f(x) = M.$$



(a) $\lim_{x \to c^+} f(x) = L$



(b) $\lim_{x \to c^-} f(x) = M$

Evaluating One-Sided Limits

Example 1

For the function

$$f(x) = \begin{cases} x+1 & \text{if } x < 2 \\ x^2 - 4 & \text{if } x \geq 2 \end{cases},$$

evaluate each of the following limits.

(a) $\lim_{x \to 2^-} f(x)$ (b) $\lim_{x \to 2^+} f(x)$

Solution:

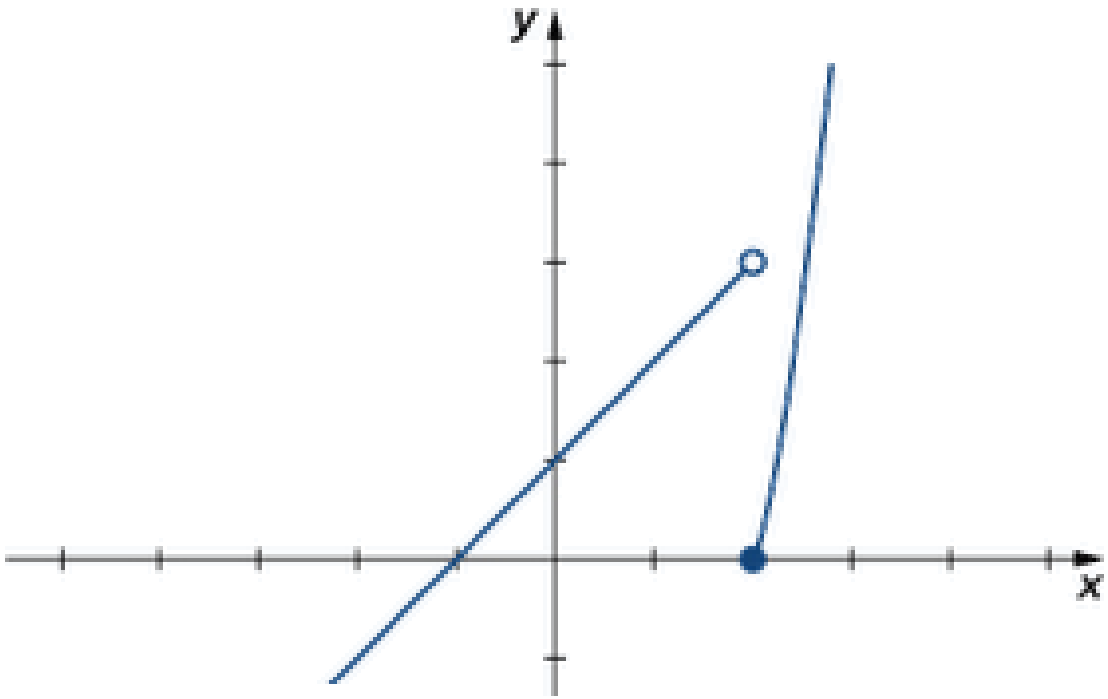
We can use tables of functional values again Table. Observe that for values of x less than 2, we use $f(x) = x + 1$ and for values of x greater than 2, we use $f(x) = x^2 - 4$.

x	$f(x) = x+1$	x	$f(x) = x^2 - 4$
1.9	2.9	2.1	0.41
1.99	2.99	2.01	0.0401
1.999	2.999	2.001	0.004001
1.9999	2.9999	2.0001	0.00040001
1.99999	2.99999	2.00001	0.0000400001

Evaluating One-Sided Limits

Example 1 Cont..

Based on this table, we can conclude that (a) $\lim_{x\to 2^-} f(x) = 3$ and (b) $\lim_{x\to 2^+} f(x) = 0$. Therefore, the (two-sided) limit of $f(x)$ does not exist at $x = 2$. The Figure below shows a graph of $f(x)$ and reinforces our conclusion about these limits.



Evaluating One-Sided Limits

Example 2

Let

$$f(x) = \begin{cases} \dfrac{x^3 - 27}{x - 3}, & \text{if } x < 3; \\ 27, & \text{if } x = 3; \\ \dfrac{x^2 + 21x - 27}{x - 3}, & \text{if } x > 3. \end{cases}$$

Find $\lim\limits_{x \to 3} f(x)$.

Solution:

Since $f(x)$ is defined by different expressions for $x < 3$ and $x > 3$, we have to find $\lim_{x \to 3^+} f(x)$, and $\lim_{x \to 3^-} f(x)$.

Evaluating One-Sided Limits

Example 2 Cont..

$$\lim_{x \to 3^+} f(x)$$

Solution:

$$\begin{aligned}
\lim_{x \to 3^+} f(x) &= \lim_{x \to 3^+} \frac{x^2 + 21x - 27}{x - 3} \\
&= \lim_{x \to 3^+} \frac{(x-3)(x+24)}{x-3} \\
&= \lim_{x \to 3^+} \frac{\cancel{(x-3)}(x+24)}{\cancel{x-3}} = 27
\end{aligned}$$

$x \to 3^+ \Rightarrow x > 3$ direct substitution gives $\left(\frac{0}{0}\right)$. Factoring $(x-3)$.

Evaluating One-Sided Limits

Example 2 Cont..

$$\lim_{x\to 3^-} f(x)$$

Solution:

$$\begin{aligned}\lim_{x\to 3^-} f(x) &= \lim_{x\to 3^-} \frac{x^3-27}{x-3}\\ &= \lim_{x\to 3^-} \frac{(x-3)(x^2+3x+9}{x-3}\\ &= \lim_{x\to 3^-} \frac{\cancel{(x-3)}(x^2+3x+9}{\cancel{x-3}} = 27\end{aligned}$$

$x \to 3^- \Rightarrow x < 3$ direct substitution gives $(\frac{0}{0})$. Factoring $(x-3)$.

Hence $\lim_{x\to 3} f(x) = 27$ Since $\lim_{x\to 3^+} f(x) = 27 = \lim_{x\to 3^-} f(x)$

Relating One-Sided and Two-Sided Limits

Theorem

Let $f(x)$ be a function defined at all values in an open interval containing c, with the possible exception of c itself, and let L be a real number. Then,

$$\lim_{x\to c} f(x) = L \iff \lim_{x\to c^-} f(x) = L \text{ and } \lim_{x\to c^+} f(x) = L$$

Example 3

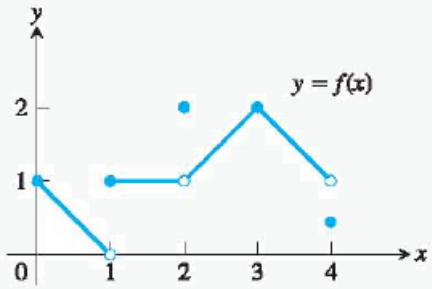


FIGURE 2.27 Graph of the function in Example 2.

EXAMPLE 2 For the function graphed in Figure 2.27,

At $x = 0$: $\lim_{x\to 0^+} f(x) = 1$,

$\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0} f(x)$ do not exist. The function is not defined to the left of $x = 0$.

At $x = 1$: $\lim_{x\to 1^-} f(x) = 0$ even though $f(1) = 1$,

$\lim_{x\to 1^+} f(x) = 1$,

$\lim_{x\to 1} f(x)$ does not exist. The right- and left-hand limits are not equal.

At $x = 2$: $\lim_{x\to 2^-} f(x) = 1$,

$\lim_{x\to 2^+} f(x) = 1$,

$\lim_{x\to 2} f(x) = 1$ even though $f(2) = 2$.

At $x = 3$: $\lim_{x\to 3^-} f(x) = \lim_{x\to 3^+} f(x) = \lim_{x\to 3} f(x) = f(3) = 2$.

At $x = 4$: $\lim_{x\to 4^-} f(x) = 1$ even though $f(4) \neq 1$,

$\lim_{x\to 4^+} f(x)$ and $\lim_{x\to 4} f(x)$ do not exist. The function is not defined to the right of $x = 4$.

At every other point c in $[0, 4]$, $f(x)$ has limit $f(c)$. ■

Limit Does Not Exist from Either Side

Example 4

Show that $\lim_{x \to 0} \sin\left(\frac{1}{x}\right)$ has no limit as x approaches 0 from either side. The Table below lists values for the function $\sin\left(\frac{1}{x}\right)$ for the given values of x.

x	$\sin\left(\frac{1}{x}\right)$	x	$\sin\left(\frac{1}{x}\right)$
-0.1	0.544021110889	0.1	-0.544021110889
-0.01	0.50636564111	0.01	-0.50636564111
-0.001	-0.8268795405312	0.001	0.8268795405312
-0.0001	0.305614388888	0.0001	-0.305614388888
-0.00001	-0.035748797987	0.00001	0.035748797987
-0.000001	0.349993504187	0.000001	-0.349993504187

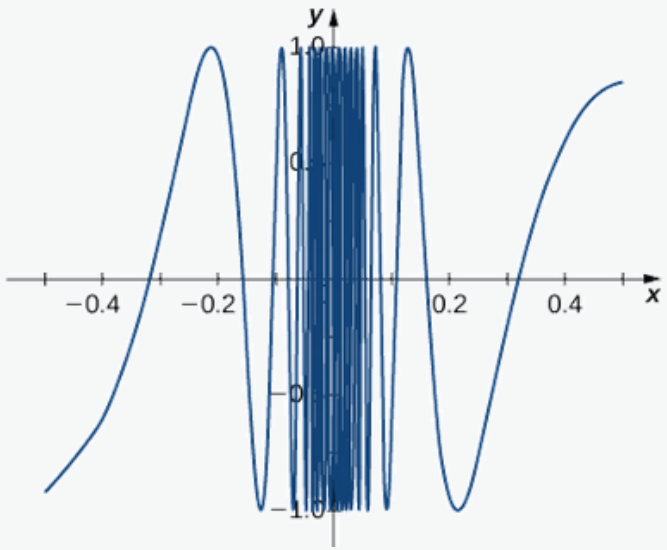
We can see that the $y-$values do not seem to approach any one single value, hence the limit does not exist. Before drawing this conclusion, let's take the following sequence of $x-$values approaching 0:

Limit Does Not Exist from Either Side

Example 4 Cont..

$$\frac{2}{\pi}, \frac{2}{3\pi}, \frac{2}{5\pi}, \frac{2}{7\pi}, \frac{2}{9\pi}, \frac{2}{11\pi}, \ldots.$$

The corresponding y-values are $1, -1, 1, -1, 1, -1, \ldots$. Hence $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$ does not exist. The graph of $f(x) = \sin\left(\frac{1}{x}\right)$ is shown in the Figure below and we can see that $\sin\left(\frac{1}{x}\right)$ oscillates between -1 and 1 as x approaches 0.

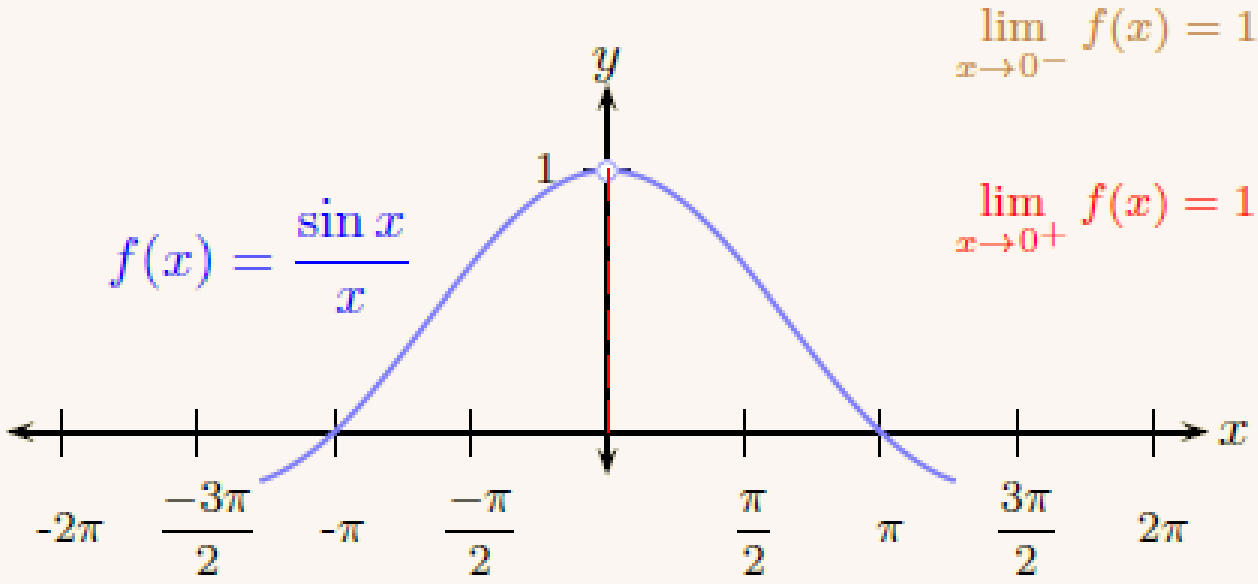


Limits Involving $\frac{\sin x}{x}$

Theorem

For x in radians,

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$



Limits Involving $\frac{\sin x}{x}$

Example 5

Evaluate $\lim_{x \to 0} \frac{\sin x}{x}$ using a table of functional values.

Solution:

We have calculated the values of $f(x) = \frac{\sin x}{x}$ for the values of x listed in Table below:

x	$\frac{\sin x}{x}$	x	$\frac{\sin x}{x}$
-0.1	0.998334166468	0.1	0.998334166468
-0.01	0.999983333417	0.01	0.999983333417
-0.001	0.999999833333	0.001	0.999999833333
-0.0001	0.999999998333	0.0001	0.999999998333

As we read down each $\frac{\sin x}{x}$ column, we see that the values in each column appear to be approaching one. Thus, it is fairly reasonable to conclude that $\lim_{x \to 0} \frac{\sin x}{x} = 1$.

Example 6

Show that (a) $\lim_{h\to 0} \dfrac{\cos h - 1}{h} = 0$ and $(b)\ \lim_{x\to 0} \dfrac{\sin 2x}{5x} = \dfrac{2}{5}$

Solution: Part (a)

(a) Using the half-angle formula $\cos h = 1 - 2\sin^2(h/2)$, we calculate

$$\begin{aligned}
\lim_{h\to 0} \frac{\cos h - 1}{h} &= \lim_{h\to 0} -\frac{2\sin^2(h/2)}{h} \\
&= -\lim_{\theta\to 0} \frac{\sin\theta}{\theta}\sin\theta \quad \text{Let } \theta = h/2 \\
&= -(1)(0) = 0
\end{aligned}$$

Example 6 Cont..

$$(b)\ \lim_{x\to 0}\frac{\sin 2x}{5x}=\frac{2}{5}$$

Solution: Part (b)

(b) Equation (1) does not apply to the original fraction. We need a $2x$ in the denominator, not a $5x$. We produce it by multiplying numerator and denominator by $2/5$:

$$\begin{aligned}\lim_{x\to 0}\frac{\sin 2x}{5x} &= \lim_{x\to 0}\frac{(2/5)\cdot \sin 2x}{(2/5)\cdot 5x}\\ &= \frac{2}{5}\lim_{x\to 0}\frac{\sin 2x}{2x}\\ &= \frac{2}{5}(1)=\frac{2}{5}\end{aligned}$$

Example 7

Evaluate the following limits:

1. $\lim_{x\to 0} \dfrac{\sin(3x)}{x}$ 2. $\lim_{x\to 0} \dfrac{\sin^3 x}{x^3}$

Solution: 1

Direct substitution will give us I.F. type $\left(\frac{0}{0}\right)$. The idea is to multiply and divide by 3, which will make the denominator the same as the argument of the sine function;

$$\begin{aligned}\lim_{x\to 0} \frac{\sin(3x)}{x} &= \lim_{x\to 0} \frac{3\sin(3x)}{3x} \\ &= 3\lim_{x\to 0} \frac{\sin(3x)}{3x} \\ &= 3(1) = 3\end{aligned}$$

Factor 3 and use the fact $\lim_{x\to 0} \dfrac{\sin(3x)}{3x} = 1$.

Example 7 Cont..

2. $\lim\limits_{x\to 0} \dfrac{\sin^3 x}{x^3}$

Solution: 2

Direct substitution will give us I.F. type $\left(\frac{0}{0}\right)$. Remember that $\frac{\sin^3 x}{x^3} = \left(\frac{\sin x}{x}\right)^3$. Now,

$$\lim_{x\to 0} \frac{\sin^3 x}{x^3} = \lim_{x\to 0} \left(\frac{\sin x}{x}\right)^3 \qquad \text{Use the power law}$$

$$= \left(\lim_{x\to 0} \frac{\sin x}{x}\right)^3 = (1)^3 = 1$$

Limits Involving $\frac{\sin x}{x}$

Example 8

Evaluate the following limits $\lim\limits_{x\to 0}\dfrac{\tan x}{x}$ and $\lim\limits_{x\to 0}\dfrac{x}{\tan x}$

Solution:

$$\lim_{x\to 0}\frac{\tan x}{x} = \lim_{x\to 0}\frac{\frac{\sin x}{\cos x}}{x}$$

Use the fact $\tan x = \dfrac{\sin x}{\cos x}$

$$= \lim_{x\to 0}\frac{\sin x}{x\cos x}$$

Use the fact $\dfrac{\frac{a}{b}}{c} = \dfrac{a}{bc}$

$$= \lim_{x\to 0}\left[\frac{\sin x}{x}\frac{1}{\cos x}\right]$$

Use the product low

$$= \left(\lim_{x\to 0}\frac{\sin x}{x}\right)\left(\lim_{x\to 0}\frac{1}{\cos x}\right)$$

$$= (1)\left(\frac{1}{\cos 0}\right) = 1$$

Use the fact $\lim\limits_{x\to 0}\dfrac{\sin x}{x}$ and direct substitution.

Example 8 Cont..

Evaluate $\lim_{x \to 0} \frac{x}{\tan x}$

Solution:

$$\lim_{x \to 0} \frac{x}{\tan x} = \lim_{x \to 0} \frac{1}{\frac{\tan x}{x}} = \frac{1}{1} = 1$$

Example 9

Evaluate the following limit: $\lim_{x\to 0} \dfrac{\sin(3x)}{\sin(4x)}$

Solution:

$$\lim_{x\to 0} \frac{\sin(3x)}{\sin(4x)} = \lim_{x\to 0} \frac{\frac{\sin(3x)}{x}}{\frac{\sin(4x)}{x}} = \lim_{x\to 0} \frac{\frac{3\sin(3x)}{3x}}{\frac{4\sin(4x)}{4x}}$$

$$= \frac{3\lim\limits_{x\to 0} \frac{\sin(3x)}{3x}}{4\lim\limits_{x\to 0} \frac{\sin(4x)}{4x}} = \frac{3(1)}{4(1)} = \frac{3}{4}.$$

Use the fact $\frac{b}{c} = \frac{\frac{b}{a}}{\frac{c}{a}}$ and divide by x. Use the quotient law.

End of Section 2.4.

Thank you.!!!